

MASTER IMPLEMENTATION SPECIFICATION

Trustworthy AI for Protein Engineering

Discovering Mutation-Space Reliability and Applying It to Industrial Carbon-Capture Enzyme Design

Version 2.0

Document Purpose

This document translates the scientific blueprint into a complete software implementation specification.

The Blueprint defines:

Why the project exists.

This document defines:

What must be built.

The purpose of the software system is to:

Measure Sparsity

↓

Measure Prediction Failure

↓

Build Reliability Framework

↓

Apply Framework To TKT

↓

Generate Industrial Candidate Rankings

The implementation must directly support the scientific goals described in the Final Project Blueprint.

System Mission

The system shall determine whether mutation-space sparsity predicts protein prediction failure and whether that information can be transformed into a practical framework for reliability-aware protein engineering.

The system must:

Compute Sparsity

Compute Antisymmetry

Compute Epistasis Error

Build Reliability Scores

Generate Industrial Fitness Scores

Evaluate Entire TKT Mutation Landscape

Rank Engineering Candidates

All outputs must be:

Reproducible

Traceable

Versioned

Scientifically Auditable

High-Level Architecture

The complete system architecture is:

Infrastructure Layer



Dataset Layer



Sparsity Framework



Experiment 1
(Discovery)



Experiment 2
(Validation)



Integrated Reliability Analysis



Experiment 3
(Application)



Reporting &
Publication Outputs

Infrastructure Layer

The infrastructure layer provides all shared functionality required by the remainder of the system.

Modules include:

Configuration Management

Logging

Checkpointing

Dataset Registry

Schema Validation

Task Management

Run Metadata

Experiment Tracking

Outputs:

logs/

checkpoints/

configs/

metadata/

Dataset Layer

The Dataset Layer manages every dataset throughout the project.

Responsibilities include:

Dataset Registration

Schema Enforcement

Version Control

Validation

Read/Write Operations

The Dataset Layer must prevent:

Missing Columns

Unexpected Columns

Invalid Data Types

Duplicate Records

Schema Drift

Every dataset used within the project must pass validation before downstream processing begins.

Sparsity Framework

The Sparsity Framework is the scientific foundation of the system.

Its purpose is to quantify how isolated a mutation is within mutation space.

The framework consists of three independent components.

Module 1 — Empirical Sparsity

Purpose:

Measure experimental mutation density.

Input:

Megascale-D data

Output:

empirical_sparsity.parquet

For each mutation:

Mutation ID

Observed Frequency

Empirical Sparsity Score

must be generated.

Module 2 — Evolutionary Sparsity

Purpose:

Measure evolutionary rarity.

Method:

ESM2

Input:

Protein Sequence

``

Output:

evolutionary_sparsity.parquet

For each mutation:

ESM Probability

Evolutionary Sparsity

must be produced.

Module 3 — Structural Sparsity

Purpose:

Measure structural isolation.

Input:

Protein Structure

SASA Calculations

Output:

structural_sparsity.parquet

For each mutation:

Residue

SASA

Structural Sparsity

must be generated.

Module 4 — Combined Sparsity

Purpose:

Generate a single sparsity metric.

Input:

Empirical Sparsity

Evolutionary Sparsity

Structural Sparsity

Output:

combined_sparsity.parquet

Fields:

Mutation

Empirical Sparsity

Evolutionary Sparsity

Structural Sparsity

Combined Sparsity

This dataset becomes the primary explanatory variable throughout the project.

Experiment 1 — Discovery

Sparsity and Thermodynamic Inconsistency

Purpose:

Determine whether sparse mutations exhibit larger thermodynamic inconsistencies.

Scientific question:

Does Combined Sparsity
predict Antisymmetry Error?

Antisymmetry Module

Inputs:

Benchmark Proteins

Selected Mutations

Prediction Models

For each mutation:

Forward calculation:

WT → MUT

Reverse calculation:

MUT → WT

must be performed.

Antisymmetry Calculation

For every mutation:

Antisymmetry Error

=

| Forward + Reverse |

must be computed.

Output:

antisymmetry_results.parquet

Fields:

Mutation

Forward $\Delta\Delta G$

Reverse $\Delta\Delta G$

Antisymmetry Error

Combined Sparsity

Discovery Analysis

Required analyses:

Correlation

Regression

Distribution Analysis

Quantile Analysis

Primary output:

Combined Sparsity

↓

Antisymmetry Error

relationship.

Experiment 2 — Validation

Sparsity and Prediction Failure

Purpose:

Determine whether sparse mutations exhibit larger real-world prediction failures.

Scientific question:

Does Combined Sparsity
predict Epistasis Error?

Epistasis Module

Inputs:

Experimental Double Mutants

Single Mutant Predictions

Double Mutant Predictions

For each pair:

Experimental Epistasis

Predicted Epistasis

must be calculated.

Epistasis Error Calculation

For every mutation pair:

Epistasis Error

=

| Predicted – Experimental |

Output:

epistasis_results.parquet

Fields:

Pair ID

Experimental Epistasis

Predicted Epistasis

Epistasis Error

Combined Sparsity

Validation Analysis

Required analyses:

Correlation

Regression

Stratified Analysis

Error Distribution Analysis

Primary output:

Combined Sparsity

↓

Epistasis Error

relationship.

Integrated Reliability Analysis

This section serves as the scientific centerpiece of the project.

Purpose:

Determine whether thermodynamic inconsistency acts as an intermediate signal between sparsity and prediction failure.

Core Relationship

The system must evaluate:

Combined Sparsity

↓

Antisymmetry Error

↓

Epistasis Error

Required outputs:

Integrated Dataset

Integrated Figures

Integrated Statistics

Output dataset:

integrated_reliability_analysis.parquet

Reliability Framework

The Reliability Framework transforms scientific discovery into an engineering tool.

Purpose:

Estimate how trustworthy a prediction is likely to be.

The framework must generate:

Reliability Score

for every evaluated mutation.

Output:

reliability_scores.parquet

Fields:

Mutation

Combined Sparsity

Reliability Score

Reliability Category

Example categories:

High Reliability

Medium Reliability

Low Reliability

Experiment 3 — Industrial Application

Reliability-Guided Mutation Engineering

Purpose:

Apply the Reliability Framework to a real engineering problem.

Selected Enzyme:

Transketolase (TKT)

Complete Mutation Landscape Generation

The system shall generate:

Every Possible
Single Amino-Acid Substitution

for the selected Transketolase sequence.

For each residue:

19 alternatives

must be generated automatically.

Output:

tkt_single_mutation_landscape.parquet

TKT Evaluation Pipeline

For every mutation:

The system shall calculate:

Evolutionary Sparsity

Structural Sparsity

Combined Sparsity

Reliability Score

Predicted Stability

Structural Metrics

Results must be merged into a unified dataset.

Output:

tkt_mutation_analysis.parquet

Active-Site Control Analysis

Purpose:

Determine whether sparsity remains informative after accounting for active-site proximity.

For every mutation:

Distance To Active Site

must be calculated.

Output:

active_site_analysis.parquet

Required analyses:

Correlation

Multivariate Regression

Stratified Comparisons

Industrial Fitness Framework

Purpose:

Generate candidate rankings.

The system must calculate:

Industrial Fitness Score

using:

Predicted Stability

Reliability Score

Evolutionary Information

Structural Information

Output:

industrial_fitness_scores.parquet

Fields:

Mutation

Reliability Score

Predicted Stability

Industrial Fitness Score

Rank

Candidate Discovery Engine

Purpose:

Identify top engineering candidates.

Output:

top_candidate_mutations.csv

Contents:

Rank

Mutation

Reliability

Industrial Fitness Score

Supporting Metrics

Figure Generation System

The system must automatically generate all figures required for:

Scientific Paper

Science Fair Board

ISEF Presentation

Required figure groups:

Discovery Figures

Validation Figures

Integrated Analysis Figures

TKT Mutation Landscape Figures

Industrial Fitness Figures

The Killer Figure

A dedicated figure-generation module shall create:

Discovery



Validation



Application

visualization.

The figure must communicate:

Megascale-D



Sparsity



Reliability



TKT Landscape



Candidate Ranking

in a single visual.

This figure becomes the flagship figure of the project.

Final Deliverables

The completed system must produce:

combined_sparsity.parquet

antisymmetry_results.parquet

epistasis_results.parquet

integrated_reliability_analysis.parquet

reliability_scores.parquet

tkc_single_mutation_landscape.parquet

tkc_mutation_analysis.parquet

active_site_analysis.parquet

industrial_fitness_scores.parquet

top_candidate_mutations.csv

along with all associated figures and tables.

System Success Criteria

The implementation is considered complete when:

✓ Sparsity Framework Operational

✓ Experiment 1 Complete

✓ Experiment 2 Complete

✓ Integrated Reliability Analysis Complete

✓ Reliability Framework Complete

✓ Full TKT Landscape Generated

✓ Active-Site Analysis Complete

✓ Industrial Fitness Framework Complete

✓ Candidate Rankings Generated

✓ All Figures Generated

✓ All Datasets Validated

✓ Results Reproducible

The resulting system must support the complete scientific objective:

Understand Prediction Failure

↓

Estimate Prediction Reliability

↓

Improve Engineering Decisions

through a fully reproducible computational framework for trustworthy AI-guided protein engineering.